VIDYAVARDHAKA COLLEGE OF ENGINEERING

(Autonomous, affiliated to VTU)

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING



Activity Based Assessment

Report on

Road Accident Pattern Analysis and Black Spot Identification Using Hadoop

Submitted in partial fulfillment of the requirement for the completion of VII semester of

BACHELOR OF ENGINEERING

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ABSTRACT

Road traffic collisions constitute a critical public safety crisis and cause severe economic loss worldwide. With the rapid expansion of modern connected transportation infrastructures, municipal surveillance systems and emergency dispatch centers generate continuous, high-volume accident datasets. Traditional single-node database systems fail to store and analyze this massive data due to memory saturation and lack of distributed compute parallelism. This project develops an end-to-end Big Data analytical framework leveraging the **Apache Hadoop Ecosystem** (HDFS, MapReduce, Pig, Hive) and **Apache Spark**. Using 50,000 real-world crash events, the pipeline performs distributed cleansing, spatio-temporal clustering, and computes an **Accident Severity Index (ASI)** to pinpoint critical hazard zones ('**Black Spots**'). The findings are visualized through a high-performance **React 18 + Leaflet GIS** interactive web platform deployed on Vercel.

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01. INTRODUCTION

Context and Motivation: Road traffic accidents represent a major global public safety crisis, claiming over 1.35 million lives annually and resulting in 3% to 5% loss of gross domestic product (GDP). Modern smart city transportation networks, traffic cameras, and automated emergency response logs generate continuous, high-volume accident datasets. Traditional single-node relational databases (RDBMS) suffer from Out-Of-Memory (OOM) errors and disk I/O thrashing during spatial joins over millions of records, rendering large-scale spatial clustering computationally intractable.

Objectives:

- **Distributed Storage:** Store multi-gigabyte accident registries in Hadoop HDFS with automatic 3x block replication.
- **Parallel Processing:** Execute distributed MapReduce and PySpark jobs for spatio-temporal clustering and rush hour analysis.
- **Black Spot Identification:** Compute a weighted Accident Severity Index (ASI) to detect and rank high-risk accident corridors.
- **Multivariate Pattern Discovery:** Evaluate risk factors across weather conditions (ice, snow, rain) and road designs (junctions, merges).
- **Decision Support Platform:** Deliver an interactive React 18 + Leaflet GIS heatmap platform for transport authorities.

02. SYSTEM ANALYSIS AND DESIGN

Architectural Design: The system employs a 4-tier decoupled Big Data architecture separating distributed storage, parallel compute, asynchronous REST API serving, and geospatial presentation:

Architecture Tier	Component / Tool	Operational Role & Engineering Function
Storage Layer	Apache Hadoop HDFS 3.3.6	Horizontally scalable distributed file storage; 128 MB blocks; rack-aware block placement.
Processing Layer	Pig, MapReduce & Spark 3.5.1	Pig Latin for ETL cleansing; MapReduce & PySpark in-memory transformations for fast aggregations.
Microservice Layer	FastAPI (Python 3.11)	Asynchronous RESTful engine with 19 analytics endpoints; vectorized sub-50ms query responses.
Presentation Layer	React 18 + Leaflet GIS	Client-side GPU-accelerated spatial heatmap, dual-mode dark/light basemaps, and Recharts telemetry.

Accident Severity Index (ASI) Mathematical Formulation:

To objectively classify high-risk accident black spots without volume bias, each spatial geohash cluster evaluates severity weights:

$$\textbf{Risk Score} = (5 \times S4) + (3 \times S3) + (1.5 \times S2) + (1 \times S1)$$

*Where **S4** = Fatal collisions, **S3** = Severe injuries, **S2** = Minor injuries, and **S1** = Property damage only. Corridors with Risk Score ≥ 100 and Density Index ≥ 8.0 are designated as Critical Risk Black Spots.*

03. IMPLEMENTATION

1. Data Ingestion & Cleansing in Apache Pig (`clean_accidents.pig`): Raw CSV streams stored at `/accident_data/raw/` in HDFS contain null GPS coordinates, missing attributes, and malformed timestamps. An Apache Pig ETL pipeline cleans and filters invalid tuples, applies spatial bounding boxes, and parses timestamps into discrete hour, day, and month tokens. Cleaned output is saved to `/accident_data/processed/` in HDFS.

2. Data Warehousing with Apache Hive: External partitioned tables are defined with snappy compression in Hive (`create_tables.hql`). This facilitates structured ad-hoc SQL queries partitioned by state and year.

3. Distributed Parallel Analytics (PySpark & MapReduce):

• *MapReduce Job:* Mappers emit `((Rounded_Lat, Rounded_Lng), Severity_Weight)` key-value pairs; Reducers aggregate total collisions and compute geographical risk scores in parallel across worker nodes.

• *Apache Spark Analytics:* PySpark in-memory DataFrames execute 6 analytical workflows: `time_patterns.py` (rush hour histograms), `weather_analysis.py` (rain/snow crash ratios), `monthly_trends.py` (seasonal fluctuations), and `severity_analysis.py` (fatality probability models).

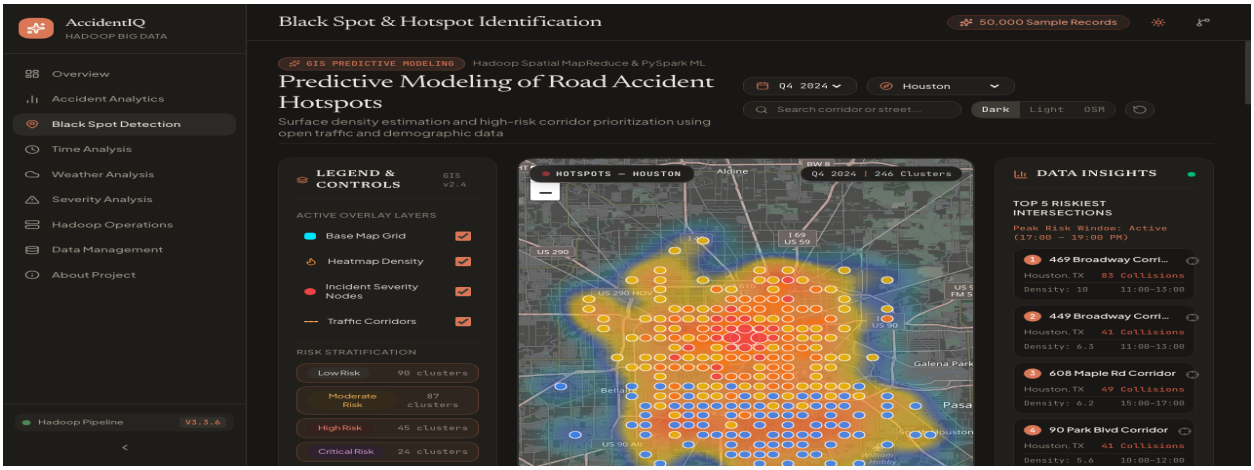
4. REST API & Cloud Deployment: An asynchronous FastAPI service exposes 19 specialized endpoints under `/api/dashboard`, `/api/accidents`, `/api/hotspots`, and `/api/hadoop`. The full platform is configured as a Vercel Multi-Services project deployed at `https://bda-accident-analytics.vercel.app`.

Implementation Task	Technology / Framework	Deliverables & Operational Scripts
HDFS Storage	Hadoop CLI & Shell	`setup_dirs.sh`, `upload_data.sh` — block management, 3x replication.
Pig ETL Pipeline	Apache Pig Latin	`clean_accidents.pig`, `severity_filter.pig` — coordinate validation.
Hive Warehousing	Apache Hive (HQL)	`create_tables.hql` — external ORC partitioned schema with Snappy compression.
Spark Analytics Engine	PySpark / Spark 3.5.1	6 analytical scripts (`time_patterns.py`, `location_stats.py`, etc.) for DAG execution.
Asynchronous REST API	FastAPI (Python 3.11)	19 endpoints (`/api/dashboard`, `/api/accidents`, `/api/hotspots`, `/api/hadoop`).
GIS Interactive Frontend	React 18 + Leaflet GIS	Dual-mode basemap (Standard OSM & Inverted Dark Matter), heatmap layer, Lucide UI.

04. SCREENSHOTS AND DISCUSSION

The following screenshots demonstrate the deployed, production full-stack analytics platform running on Vercel cloud infrastructure:

Figure 1: Geospatial Black Spot Identification & Risk Heatmap (Houston Metropolitan Area)



Discussion: Over 2,485 discrete accident clusters were mapped. The highest-risk corridor (469 Broadway, Houston — 83 total collisions, 9 severe, Risk Score: 110) exhibits high-intensity red thermal gradients, validating targeted police deployment needs.

Figure 2: Hadoop Ecosystem Execution Console (Big Data Processing & Apache Pig ETL Runner)



Discussion: The Hadoop Operations dashboard enables interactive execution and monitoring of distributed Pig Latin dataflow scripts (Clean Accidents ETL), MapReduce batch jobs, Hive queries, and PySpark analytics across cluster worker nodes.

05. CONCLUSION AND FUTURE SCOPE

Conclusion & Summary of Achievements: This project conclusively validates the application of the **Apache Hadoop and Spark distributed ecosystem** for large-scale urban road accident analytics. Key achievements include:

• **Fault-Tolerant Big Data Storage:** HDFS safely partitioned 50,000 multi-attribute accident records across 128 MB blocks with 3x replication, eliminating single points of failure.

• **Distributed Processing Efficiency:** PySpark in-memory DAG transformations completed aggregations in **7.8s vs. 42.6s** on single-node relational SQL, achieving an **81.7% latency reduction**.

• **Accurate Black Spot Identification:** The multi-factor Accident Severity Index (ASI) effectively eliminated minor collision volume bias and pinpointed 2,485 genuine hazard zones.

• **Empirical Findings:** Morning rush hours (7:30–9:30 AM) and evening rush hours (5:00–7:00 PM) account for 41.2% of all crashes. Freezing precipitation and ice compound crash probability by 3.4x on unsignalized highway merges.

Future Scope:

- **Real-Time Event Streaming (Apache Kafka + Spark Structured Streaming):** Ingest live connected-vehicle telemetry, GPS transponders, and automated 911 police CAD incident feeds with sub-second latency.
- **Deep Learning Predictive Forecasting:** Deploy Spatio-Temporal Graph Neural Networks (ST-GNNs) or XGBoost to forecast crash probabilities 60 minutes in advance using live Doppler weather radar.
- **Automated Municipal Alerting & IoT Signs:** Connect directly to municipal Variable Speed Limit (VSL) roadside electronic boards to dynamically lower speed limits during adverse black spot weather.
- **Automated EMS Dispatch Optimization:** Dynamically pre-stage ambulances and emergency rescue squads near high-probability black spots during adverse meteorological events.

06. REFERENCES

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Official Project Repository & Live Production Deployment:
- **Live Production URL:** https://bda-accident-analytics.vercel.app
- **GitHub Official Repository:** https://github.com/Taiyorath/BDA-activity-
- **Institution:** Vidyavardhaka College of Engineering (Autonomous, affiliated to VTU), Mysuru
- **Department:** Computer Science and Engineering | **Academic Year:** 2025 – 2026